

## **Structured Training Program: Wi-Fi**

### *Module 2: Introduction and History of Wi-Fi*

1. Brief about SplitMAC architecture and explain how it improves AP's performance.

SplitMAC is an architecture design for wireless networks that divide IEEE 802.11 protocol tasks between a Lightweight Access Point (LAP) and a Wireless LAN controller (WLC).

The AP controls realtime, time-sensitive tasks (beaconing, buffering, encryption). The WLC handles non-realtime centralized tasks (authentication, association with APs and maintain security policy).

SplitMAC improves the AP's performance in following ways,

- a) Avoids APs from using valuable CPU cycles on tasks such as authentication, association and roaming management etc.,
- b) Instead of each AP independently making, potentially conflicting, radio resource management (RRM) decisions, the WLC analyzes data from all APs to optimize channel selection and transmission power.
- c) Central control avoids interference and noise across the entire network.
- d) The controller handles client roaming, allowing for faster re-association and maintaining active sessions (like voice calls) as users move between APs.
- e) APs can be deployed without individual configuration. They connect to the WLC, download necessary firmware, and receive configuration automatically, speeding up deployment
- f) The WLC can monitor client load on each AP and actively suggest that clients connect to a less congested AP, improving overall network performance.

2. Describe the CAPWAP, explain the flow between AP and controller.

CAPWAP (Control and Provisioning of Wireless Access Points) is an IEEE standard protocol allowing a Wireless LAN Controller (WLC) to manage lightweight Access Points (APs). It separates the control plane (management, config) from the data plane (client traffic), establishing secure, encrypted UDP tunnels (Port 5246 for control, 5247 for data) to enable centralized management,

The CAPWAP consists of the following processes,

- Discover Process (AP Discovery through L2 Broadcast, L3 Broadcast, DHCP Option 43, DNS, Priming, Static, OTAP)
- DTLS handshake
- Join Process (AP joins WLC through Primary/Secondary/Tertiary/Global Backup WLC, Master Controller Flag, Least Load WLC)
- Configuration downloaded to AP (config state)
- Data & control traffic exchanged (Run state)

### Flow Between AP and Controller (AP Join Process)

| Step | Phase                      | Description                                                                                                |
|------|----------------------------|------------------------------------------------------------------------------------------------------------|
| 1    | AP Boot with DHCP          | AP powers on and gets IP address via DHCP. DHCP Option 43 may point to WLC IP.                             |
| 2    | WLC Discovery              | AP sends CAPWAP Discovery Request (broadcast/multicast or using DNS 'CISCO-CAPWAP-CONTROLLER').            |
| 3    | WLC Response               | WLC replies with CAPWAP Discovery Response including its IP.                                               |
| 4    | DTLS Handshake             | AP and WLC establish a secure DTLS (Datagram TLS) session for encrypted control communication.             |
| 5    | Join Request/Response      | AP sends CAPWAP Join Request; WLC authenticates AP (via certificate/MAC) and sends Join Response.          |
| 6    | Configuration              | WLC pushes configuration (SSID, VLAN, security policies, RF settings) to the AP via CAPWAP Control tunnel. |
| 7    | Image Download (if needed) | If the AP firmware is outdated, WLC downloads the correct firmware over CAPWAP.                            |
| 8    | Run State                  | AP enters Run State. CAPWAP Control tunnel keeps alive. Client data flows through CAPWAP Data tunnel.      |

3. Where is the CAPWAP fits in OSI model, what is the two tunnels in CAPWAP and its purpose.

CAPWAP primarily operates at Layer 4 (Transport Layer), as it uses UDP for transport. However, it is an encapsulation protocol that carries 802.11 management and data frames between the AP and WLC, often described as spanning across Layer 2 and Layer 3, and generally functions over IP networks.

#### Two tunnels in CAPWAP:

CAPWAP establishes two distinct logical tunnels over a single, secure session to separate management traffic from user data traffic.

##### i. Control Tunnel

Manages the communication between the WLC and the AP, including AP discovery, join requests, firmware upgrades, configuration delivery, and status monitoring. Uses UDP port 5246. By default, this channel is secured using DTLS

(Datagram Transport Layer Security) to encrypt configuration and management commands.

## ii. Data Tunnel

Encapsulates and forwards user data (client traffic) between the AP and the WLC in centralized forwarding scenarios. It allows client data to travel from the AP to the WLC for centralized policy application. Uses UDP port 5247. Encryption for this tunnel is typically optional (DTLS data protection can be applied).

### 4. What is the difference between Lightweight APs and cloud-based APs ?

In a Lightweight AP architecture, the intelligence is centralized in a physical or virtual controller (WLC) on your local network. In a Cloud-based architecture, the management and control functions are hosted in a vendor's data center, accessible via the internet.

| <b>Feature</b>             | <b>Lightweight APs (Cisco WLC/CAPWAP)</b>                         | <b>Cloud-Based APs (Cisco Meraki, Aruba Central)</b>                           |
|----------------------------|-------------------------------------------------------------------|--------------------------------------------------------------------------------|
| <b>Management</b>          | Centralized via an on-premise Wireless LAN Controller (WLC).      | Centralized via a Web Dashboard hosted by the vendor.                          |
| <b>Protocol</b>            | Uses CAPWAP to communicate with the controller.                   | Uses HTTPS/TLS to communicate with the cloud management platform.              |
| <b>Data Plane</b>          | Traffic is often "tunneled" back to the WLC (Central Switching).  | Traffic is usually switched locally at the AP (Local Breakout).                |
| <b>Internet Dependency</b> | Can function fully if the internet goes down (local control).     | Management is lost if the internet goes down (APs usually still pass traffic). |
| <b>Scalability</b>         | Limited by the hardware capacity of the physical WLC.             | Highly scalable; add as many APs as needed without hardware limits.            |
| <b>Cost Model</b>          | High upfront hardware cost (CapEx) for the controller.            | Lower upfront cost but requires ongoing licenses (OpEx).                       |
| <b>Deployment</b>          | Requires manual configuration of VLANs, Trunks, and WLC settings. | Zero-Touch Provisioning (ZTP); APs pull config once they reach the cloud.      |

5. How the CAPWAP tunnel is maintained between AP and controller?

CAPWAP tunnel is maintained between an Access Point (AP) and Wireless LAN Controller (WLC) through continuous, bidirectional heartbeat mechanisms. The control tunnel (UDP 5246) uses periodic Echo Request/Response packets, while the data tunnel (UDP 5247) uses Keepalive packets, ensuring session integrity and prompt detection of disconnection.

6. What is the difference between sniffer and monitor mode, use case of each mode.

In wireless networking, both Sniffer and Monitor modes are used for troubleshooting and analysis, but they differ significantly in how the data is collected and where the AP is in the network hierarchy.

| Feature              | Sniffer Mode                                                                          | Monitor Mode / Rogue Detector Mode                                              |
|----------------------|---------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|
| <b>Primary Goal</b>  | Packet Capture, grabbing raw 802.11 frames for deep analysis in tools like Wireshark. | Network Health, Security monitoring and location tracking.                      |
| <b>Radio Status</b>  | Dedicated to capturing traffic on a specific channel. It does not serve clients.      | Dedicated to scanning all channels. It does not serve clients.                  |
| <b>Data Delivery</b> | Forwards captured packets to a specific IP address (your laptop/server) via the WLC.  | Reports data (like Rogue APs or interference) to the WLC and management system. |
| <b>Scope</b>         | Narrow focus (one channel at a time).                                                 | Broad focus (cycles through all channels).                                      |

7. If WLC depends on WAN, which AP mode is best for local network and how ?

When a Wireless LAN Controller (WLC) is located across a WAN, FlexConnect mode (H-REAP or REAP) is the best Enterprise Access Point (AP) mode for a local network. It allows APs to locally switch traffic to the local network and authenticate users locally, keeping the network functional even if the WAN link to the central WLC fails.

Flexconnect is best because,

- FlexConnect allows APs to connect to a central WLC, but intelligently switches client data traffic directly onto the local network switch, rather than tunneling it back through the WAN.
- Even with the WAN down, APs can maintain connectivity for existing clients and allow new clients to authenticate locally.
- By switching data locally, latency is significantly reduced, which is crucial for applications like voice and video.
- Despite operating locally, APs still receive management and configuration from the central WLC when the WAN is active.

8. What are the challenges if deploying autonomous APs (more than 50) in large network like university?

Deploying a large number of Autonomous (Standalone) Access Points in a campus environment like a university presents significant operational and technical hurdles. Unlike Lightweight APs that use the CAPWAP protocol for centralized management, each Autonomous AP acts as an independent node.

The five primary challenges are:

1. Management and Scalability (High Administrative Overhead)

- **Manual Configuration:** There is no centralized "brain." Any change to a configuration (such as changing an SSID name or updating a security key) must be performed on every AP individually via CLI or a local web interface.
- **Firmware Management:** Upgrading software becomes a logistical nightmare. If an organization has 50+ APs, the IT staff must manually push updates 50+ times, increasing the risk of version mismatch and downtime.

2. Lack of Centralized Radio Resource Management (RRM)

- **Co-Channel Interference (CCI):** Autonomous APs cannot communicate with one another to coordinate channel selection. In a dense environment, multiple APs may choose the same channel, leading to massive interference and degraded performance.
- **Static Power Levels:** These APs cannot automatically adjust their transmit power to compensate for a failed neighbor or to reduce "cell overlap," resulting in "dead zones" or inefficient signal propagation.

3. Inefficient Roaming (Client "Stickiness")

- **No Centralized Handover:** In a controller-based system, the WLC facilitates fast roaming. In an autonomous setup, the decision to roam is left entirely to the client device.
- **Re-authentication Latency:** Because there is no central session state, a student moving between buildings may experience a drop in connection as the device is forced to perform a full re-authentication with every new AP it encounters.

4. Security and Rogue Detection Gaps

- **No Global Visibility:** Autonomous APs cannot effectively detect "Rogue APs" (unauthorized hotspots) across the entire network. There is no central dashboard to alert administrators of security threats in real-time.
- **Complex VLAN Mapping:** To support multiple user groups (Students, Faculty, Guests), the network admin must manually configure 802.1Q trunks on every individual switch port connected to an AP, making the physical infrastructure harder to maintain.

### 5. Increased Operational Expenditure (OpEx)

While autonomous APs avoid the upfront cost of a Wireless LAN Controller (WLC) license, the cost of labor to troubleshoot, monitor, and maintain 50+ disparate devices far outweighs the initial savings.

9. What happens on a wireless client connected to Lightweight AP in local mode if WLC goes down?

When a Cisco Lightweight Access Point (LAP) in Local Mode loses its connection to the Wireless LAN Controller (WLC), all connected wireless clients lose connectivity. The AP continues broadcasting SSIDs but cannot process traffic, as Local Mode relies on the WLC for authentication, encryption keys, and bridging data to the wired network.

The key characteristics are,

- Loss of Connectivity: Existing clients will not be able to pass traffic and will eventually be disconnected.
- No New Authentications: New clients cannot associate or authenticate with the network.
- AP Behavior: The LAP is "lightweight," meaning it cannot function independently without the WLC's control plane.
- Reconnection: The AP will enter an "AP Fallback" state, attempting to reconnect to the primary WLC or a configured standby WLC.
- Difference from FlexConnect: Unlike FlexConnect, which can switch to standalone mode, Local Mode APs rely on central switching, making them entirely dependent on WLC availability.